LATIHAN KOMPREHENSIF PEMROGRAMAN DATABASE

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MATKUL : PEMROGRAMAN DATABASE

A. *Membuat sebuah database dengan nama* **db_electronics_store**

```
CREATE DATABASE db_electronics_store;
USE db_electronics_store;
```

B. ***1. TABLE customers***

```
CREATE TABLE customers (
  -> customer_id INT AUTO_INCREMENT PRIMARY KEY,
  -> customer_name VARCHAR(100),
  -> email VARCHAR(100),
  -> phone_number VARCHAR(20)
  -> );
```

2. ***TABLE product_categories***

```
CREATE TABLE product_categories (

  -> category_name VARCHAR(50)
  -> category_id INT AUTO_INCREMENT PRIMARY KEY,
  -> );
```

3. ***TABLE products***

```
CREATE TABLE products (

-> product_id INT AUTO_INCREMENT PRIMARY KEY,
-> product_name VARCHAR(100),
-> price DECIMAL(10,2),
-> stock INT,
-> category_id INT,
-> FOREIGN KEY (category_id) REFERENCES  product_categories(category_id)
-> );
```

4. TABLE transactions

```
CREATE TABLE transactions (

-> transaction_id INT AUTO_INCREMENT PRIMARY KEY,
-> customer_id INT,
-> transaction_date DATE,
-> FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
-> );
```

5. *TABLE transaction_details*
CREATE TABLE transaction_details (
-> detail_id INT AUTO_INCREMENT PRIMARY KEY,
-> transaction_id INT,
-> product_id INT,
-> quantity INT,
-> subtotal DECIMAL(10,2),
-> FOREIGN KEY (transaction_id) REFERENCES transactions(transaction_id),
-> FOREIGN KEY (product_id) REFERENCES products(product_id)
->);

C. *Membuat table customer minimal 5 data*

1. customer
INSERT INTO customers (customer_name, email, phone_number) VALUES

-> ('Dewi Lestari', 'dewi@gmail.com', '081234567891'),
-> ('Budi Santoso', 'budi@gmail.com', '081234567892'),
-> ('Ani Rahma', 'ani@gmail.com', '081234567893'),
-> ('Tono Nugroho', 'tono@gmail.com', '081234567894'),
-> ('Siti Aminah', 'siti@gmail.com', '081234567895');

2. *product_categories*
INSERT INTO product_categories (category_name) VALUES
-> ('Laptops'),
-> ('Smartphones'),
-> ('Accessories'),
-> ('Television');

3. *products*
INSERT INTO products (product_name, price, stock, category_id) VALUES
-> ('Acer Aspire 5 Laptop', 7500000, 10, 1),
-> ('Samsung Galaxy A51', 4000000, 15, 2),
-> ('Logitech M185 Mouse', 150000, 30, 3),
-> ('USB Type-C Cable', 50000, 50, 3),
-> ('LG 43-inch TV', 4500000, 5, 4);

4. *transactions*
INSERT INTO transactions (customer_id, transaction_date) VALUES
-> (1, '2024-06-01'),
-> (2, '2024-06-01'),
-> (3, '2024-06-03');

5. trasnsaction_details

```
INSERT INTO transaction_details (transaction_id, product_id, quantity, subtotal) VALUES
   -> (1, 1, 1, 7500000),
   -> (1, 3, 2, 300000),
   -> (2, 2, 1, 4000000),
   -> (3, 3, 1, 150000),
   -> (3, 5, 1, 4500000);
```

D. Bebebrapa Perubahan Data

1). Mengubah nama produk acer menjadi laptop

```
UPDATE products
-> SET product_name = 'Laptop Asus Zenbook'
-> WHERE product_name = 'Acer Aspire 5 Laptop';
```

2). Mengubah stock Logitech menjadi 20

```
UPDATE products
   -> SET stock = 20
   -> WHERE product_name = 'Logitech M185 Mouse';
```

3). Mengubah nomor hp Ani Rahma

```
UPDATE customers
 -> SET phone_number = '081943658755'
 -> WHERE customer_name = 'Ani Rahma';
```

4). Hapus customer Tono Nugroho

```
DELETE FROM customers
-> WHERE customer_name = 'Tono Nugroho';
```

5). Hapus Samsung dari tabel products

```
DELETE FROM transaction_details WHERE product_id = 2;
DELETE FROM products WHERE product_id = 2;
SELECT * FROM products WHERE product_id = 2;
```

E. Membuat Informasi

1). Daftar produk lengkap: nama produk dan nama kategori

```
SELECT products.product_name, product_categories.category_name
-> FROM
-> (products JOIN product_categories ON products.category_id =
product_categories.category_id);
```

2). Daftar transaksi lengkap: nama pelanggan dan total belanja

```
SELECT customers.customer_name,
-> transaction_details.subtotal
-> FROM transactions
->  JOIN customers ON transactions.customer_id = customers.customer_id
->JOIN transaction_details ON transaction_details.transaction_id =
transactions.transaction_id;
```

3). Detail transaksi: nama pelanggan, nama produk, jumlah, subtotal, tanggal transaksi

```
SELECT customers.customer_name,
-> products.product_name,
-> transaction_details.quantity,
-> transaction_details.subtotal,
-> transactions.transaction_date
-> FROM
-> transactions
-> JOIN transaction_details ON transactions.transaction_id =
transaction_details.transaction_id
-> JOIN products ON transaction_details.product_id = products.product_id
-> JOIN customers ON transactions.customer_id = customers.customer_id;
```

4). Daftar produk yang belum pernah dibeli

```
SELECT products.product_id,
-> products.product_name
-> FROM products
-> WHERE NOT EXISTS (
-> SELECT 1 FROM transaction_details
-> WHERE transaction_details.product_id = products.product_id);
```

5). Daftar pelanggan yang sudah pernah bertransaksi

```
SELECT DISTINCT
-> customers.customer_name
-> FROM transactions
-> JOIN customers ON transactions.customer_id = customers.customer_id;
```

F. Menampilkan

1). Jumlah transaksi yang terjadi

```
SELECT COUNT(*) AS total_transactions
-> FROM transactions;
```

2). Total pendapatan per hari

```
SELECT transaction_date,
-> SUM(subtotal) as pendapatan
-> FROM transactions
-> JOIN transaction_details ON transactions.transaction_id =
transaction_details.transaction_id
-> GROUP BY transactions.transaction_date;
```

3). Rata-rata pembelian per pelanggan

```
SELECT customer_name,
-> AVG(transaction_details.subtotal) AS rata_rata_pembelian
-> FROM customers
-> JOIN transactions ON customers.customer_id = transactions.customer_id
-> JOIN transaction_details ON transactions.transaction_id =
transaction_details.transaction_id
-> GROUP BY customer_name;
```

4). Produk dengan harga tertinggi dan terendah

```
a. SELECT product_name, price
-> FROM products
-> ORDER BY price;
```

```
b. SELECT product_name, price
-> FROM products
-> ORDER BY price desc
-> LIMIT 1;
```

```
c. SELECT product_name, price
-> FROM products
-> ORDER BY price asc
-> LIMIT 1;
```

5). Total produk yang masih tersedia stok-nya

```
SELECT product_name, price, stock
-> FROM products
-> WHERE stock > 0
-> GROUP BY stock;
```

G. Membuat VIEW view_transaksi_lengkap berisi: customer_name, transaction_date, product_name, quantity, dan subtotal

```
SELECT
-> customers.customer_name,
-> transactions.transaction_date,
-> products.product_name,
-> transaction_details.quantity,
-> transaction_details.subtotal
-> FROM transactions
-> JOIN customers ON transactions.customer_id = customers.customer_id
-> JOIN transaction_details ON transactions.transaction_id = transaction_details.transaction_id
-> JOIN products ON transaction_details.product_id = products.product_id;
```